



SYDNEY CAMPUS

PRT661 — Data Science Practice

Assignment I — Project Proposal

Real-Time Wildfire Risk Tracking and Prediction System for Australia

SUBMITTED BY

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1 Project Overview and Objectives

1.1 Project Overview

The **Real-Time Wildfire Risk Tracking and Prediction System for Australia** is a highly interactive, online web-based system that has been created to help in tracking the fire risks that may arise and predicting the fire risk rating for seven days ahead in Australia. The system integrates real-time satellite data with high-resolution numerical weather prediction models to offer predictive insight to support emergency services, regional planners, and local communities within seconds.

1.2 Core Objectives

Asynchronous Data Ingestion Pipeline: Create low-latency Python REST API connections for pulling active thermal hotspots from NASA FIRMS, seven-day meteorological forecast vectors from Open-Meteo, and regional air quality data.

Feature Engineering & Automated ETL: Streamline the data streams, compute the great circle Haversine distance for the radial spatial buffer (10-50km), convert timestamp to Australian Eastern Standard Time (AEST), and generate dead fine fuel moisture indices.

Hybrid Analytics Engine: Conjoin deterministic physics, McArthur FFDI (Forest Fire Danger Index), and supervised learning classification (XGBoost/Random Forest) for the analysis of fast-moving fire probabilities.

Persistent Architecture & Offline Fallback: Use a 3NF normalised **SQLite database** (wildfire_db.db) to store API response cache, query logs by users, predictions, and offline fallbacks.

Spatial & Temporal Dashboard UI: Implement an interactive Streamlit web application with the help of Folium GIS mapping tools for hotspot detection along with Plotly timelines showing hourly fire risk.

2 Theme Justification

This project explicitly aligns with Theme 2: Predictive Analytics and Forecasting. Rather than simply displaying historical fire events, the proposed system forecasts future fire-weather conditions and derives continuous fire-danger outcomes across a 168-hour (7-day) horizon (Noble, Bary, & Gill, 1980; Open-Meteo, n.d.) .

The project also incorporates the major data science pillars required by the unit. Data Acquisition is addressed through Open-Meteo and NASA FIRMS APIs; Data Storage through SQLite; Data Processing through Pandas and NumPy; Analytics and Predictive Modelling through the McArthur fire-danger formulation and continuous prediction; and Visualization through Streamlit, Folium and Plotly.

This makes the project a data science practice project rather than a simple software dashboard. The system must acquire heterogeneous data, validate it, transform it into meaningful analytical variables, apply domain-specific mathematical modelling, communicate uncertainty, and present results in a form that users can interpret.

3 Problem Statement

3.1 Problem Statement

Wildfire risk monitoring across regional and peri-urban Australia presents three primary technical and operational limitations:

1. **Coarse Spatial Resolution:** Regional synoptic fire danger ratings are computed across broad geographical zones, masking micro-climatic variations, localized wind funneling, and fine fuel dryness differences.
2. **Disconnected Sensor Telemetry:** Active thermal satellite detections (NASA VIIRS/MODIS) are often presented on static maps without immediate integration with local weather vectors or fine fuel moisture indices.
3. **Static Empirical Indices:** Standard empirical formulas like the McArthur FFDI capture general atmospheric fire potential but lack non-linear predictive capabilities for active, fast-moving fire fronts and satellite Fire Radiative Power (FRP) metrics.

3.2 Proposed Hybrid Solution

To overcome these limitations, the system adopts a **Hybrid AI & Physics Architecture**:

- **Physics Baseline Layer:** FFDI, Equilibrium Moisture Content (EMC), and Rate of Spread (ROC) are calculated to maintain transparency in the calculation process with regards to the Australian emergency management standards.
- **Machine Learning Engine:** Operates a tree-based ensemble model classifier (`fire_model.pkl`), which takes satellite-derived fire radiative power (FRP, brightness) with weather variables as inputs and produces precise probabilities for fast-spread fires.

4 Target Users and Expected Value

- **Emergency management and disaster response analysts:** For assessing local fire weather situations and possible ignition points.
- **Land and forestry managers:** For determining times of increased fire risk and planning purposes.
- **Researchers and students:** For analysing the relationship between meteorological conditions, satellite fire observations and calculated fire-danger indicators.
- **Academic and demonstration users:** for evaluating an end-to-end GeoAI decision-support workflow.

The expected value is the integration of fragmented environmental data into a single analytical interface. The prototype is intended to support investigation and decision-making, not to replace official warnings or emergency-management systems.

5 Initial System Architecture

The initial proposed system is designed as a six-layer data pipeline architecture. The pipeline extracts the data from the external APIs that provide satellite hotspot data and meteorological information; a processing layer validates and transforms the datasets, the analytics layers performance the FFDI calculations and predict the outcome, then stores them in a lightweight database for persist runtime artifacts and finally presenting the results through a dashboard.

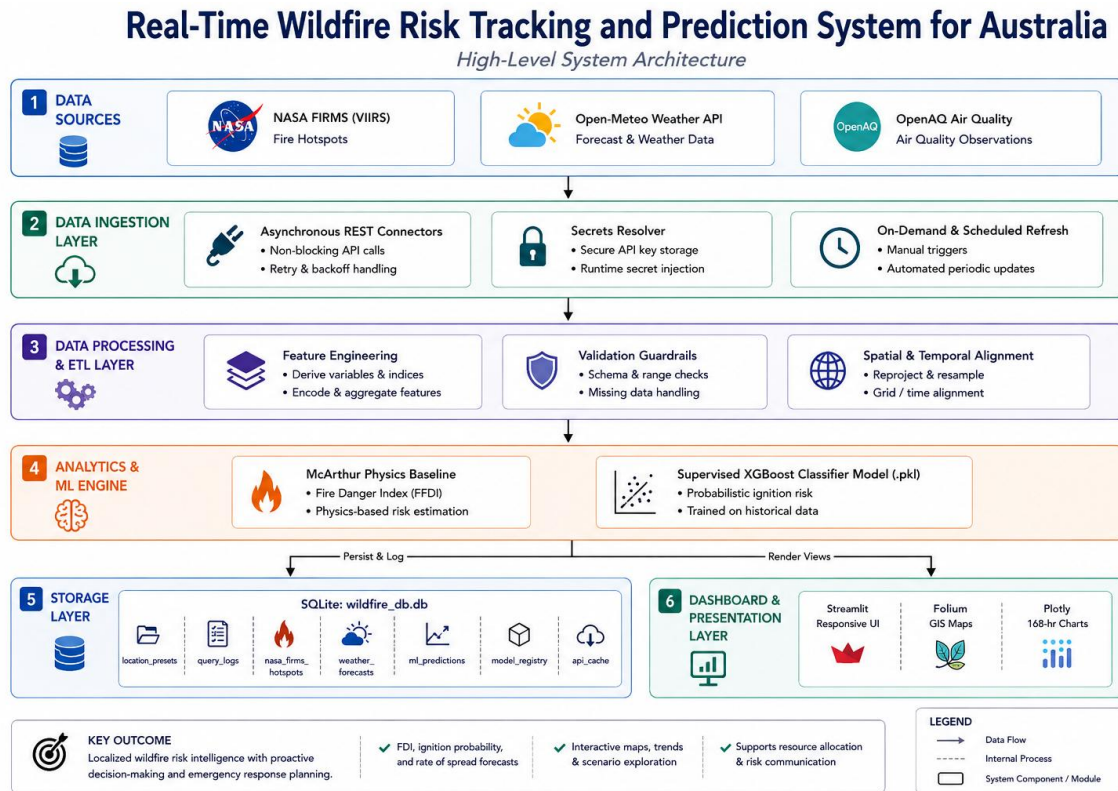


Figure 1 Initial High Level Architecture Diagram for Real-Time Wildfire Risk Tracking and Prediction System for Australia

5.1 Data Sources and Ingestion Layer

The system ingests real-time and forecast environmental data from external APIs:

- NASA FIRMS (Thermal Hotspots):** It collects active fire satellite observations from VIIRS SNPP (375m). These 15 parameters are extracted by the ingestion engine, which include latitude, longitude, brightness, scan, track, acq_date, acq_time, satellite, instrument, confidence, version, bright_t31, frp, daynight, and type.
- Open-Meteo Weather API:** It fetches 7-day meteorological forecast information, which includes 2m air temperature, relative humidity, wind speed, wind direction, and surface precipitation.

- **Air Quality API:** It provides particulate metrics for local air quality background and atmospheric smoke verification.

5.2 Data processing and ETL Pipeline

Raw datasets undergo four transformation steps to ensure data quality and integrity before modeling:

- **Validation:** It confirms the coordinates fall within the Australia and filters out low-confidence satellite noise.
- **Cleaning and Standardization:** Standardizes units, converts dates and UTC to AEST time and filters to focus on vegetation fires rather than industrial heat sources.
- **Spatial processing:** It calculates radial boundaries filtering around target coordinates.
- **Temporal Alignment:** It aligns weather time-series into continuous hourly steps with hotspot detection timestamps.

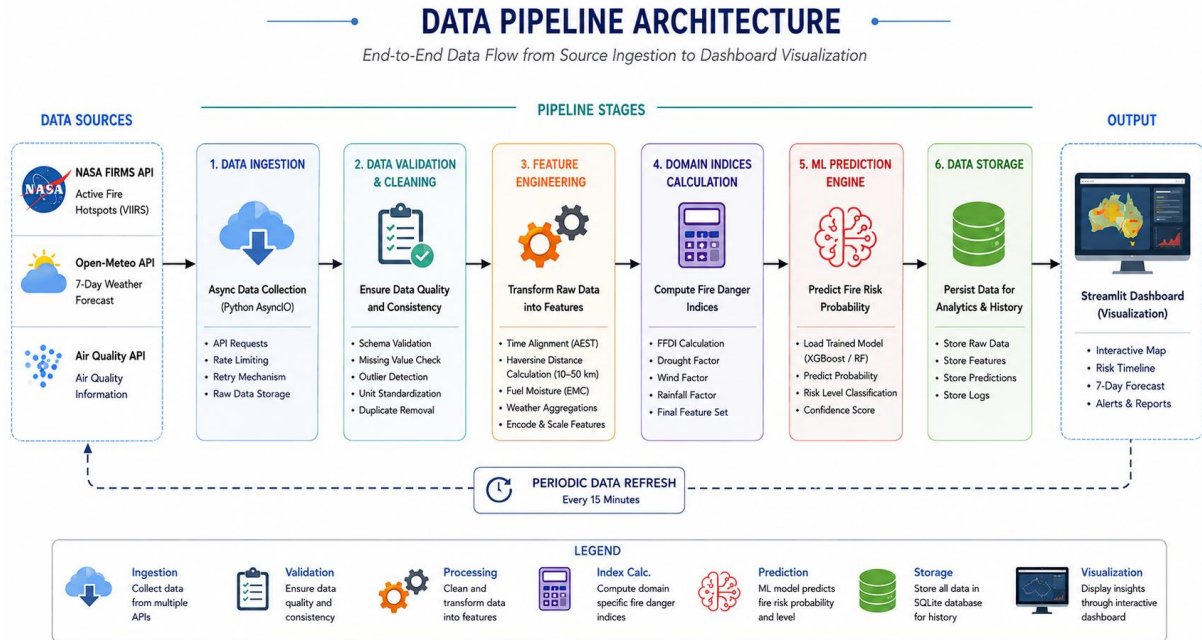


Figure 2 Data Pipeline Architecture

5.3 Analytics and Fire Danger Engine

The analytics engine applies domain-specific mathematical formulas to estimate real-time and forecast fire risks.

McArthur Forest Fire Danger Index (FFDI)

$$FFDI = 2.0 \cdot \exp(-0.450 + 0.987 \cdot \ln(DF) - 0.0345 \cdot RH + 0.0338 \cdot T + 0.0234 \cdot V)$$

where:

- DF = Drought Factor (0 - 10)
- T = Air temperature (°C)
- RH= Relative humidity (%)
- V = Wind Speed (Km/h)

Feature Engineering

- **Fire Danger Classification:** It classifies continuous FFDI values into Australian Standard operational rating:
 - Low/Moderate: $FFDI < 12$
 - High: $12 \leq FFDI < 24$
 - Extreme: $24 \leq FFDI < 50$
 - Catastrophic: $FFDI \geq 50$
- **Ignition Probability:** It estimates ignition risk percentages based on fuel moisture equilibrium (EMC) and heat inputs.

$$\text{For } RH < 10\%: EMC = 0.03229 + 0.281073 \cdot RH - 0.000578 \cdot T \cdot RH$$

$$\text{For } 10\% \leq RH \leq 50\%: EMC = 2.2272 + 0.1601 \cdot RH - 0.01478 \cdot T$$

- **Rate of Spread:** It estimates the fire front speed using fuel loading, wind speed, and active satellite fire Radiative Power (FRP).

$$ROS = 0.0012 \cdot FFDI \cdot w \text{ (Km/h)}$$

Where w is fuel loading (t/ha)

5.4 Machine Learning Engine Architecture

- **Model Type:** XGBoost / Random Forest Classifier pre-compiled and saved to fire_model.pkl.
- **Target (y):** Binary rapid fire spread event (1 = Rapid Spread / High Severity, 0 = Contained / Low Risk).
- **Input Feature Matrix (X):** [brightness, bright_t31, frp, scan, track, confidence_encoded, temp_2m, rh_2m, wind_speed_10m, ffdi_value, emc_value]
- **Evaluation Metrics:** Evaluated offline via 5-fold cross-validation targeting high Precision, Recall, F_1 -score, and ROC-AUC.

5.5 Local Storage

A lightweight SQLite database serves as the system's persistence layer.

5.6 Presentation Layer

The dashboard is built using Streamlit, integrating two visual modules:

- **Folium GIS Map Engine:** It renders interactive maps, showing active hotspots that are colour code by confidence and FRP, radical search boundaries, and spatial risk layers.
- **Plotly Charting Engine:** It displays 7-days hourly timeline charts tracking expected changes in hourly FFDI, ambient temperature, relative humidity, and ignition probability.

6 Workflow Plan

The workflow follows a four-phase agile pipeline, structure around iterative agile development principles (Beck, et al., 2001), and designed for the initial four-week delivery timeline.

6.1 Data Ingestion

Asynchronous REST calls will retrieve meteorological vectors, including temperature, wind and humidity from Open-Meteo, together with NASA FIRMS satellite thermal hotspot information within user-defined geographic search areas (NASA Fire Information for Resource Management System, n.d.; Open-Meteo, n.d.). Retrieved data will be validated and transformed into a consistent internal representation.

6.2 Analytics and Modelling

Validated weather features will be passed to the McArthur FFDI mathematical formulation to derive interpretable fire-danger ratings. The system will communicate categories such as Low, Moderate, High, Extreme and Catastrophic, alongside numerical outputs where supported by the implemented formulation (Noble, Bary, & Gill, 1980). Ignition probability will be treated as a predictive/derived indicator and clearly documented rather than presented as a guaranteed probability of fire.

6.3 Data Storage

A zero-configuration local SQLite database, wildfire_demo.db, will store query logs, regional preset coordinates and cached information to support low-latency operation and resilience during demonstrations (SQLite, n.d.).

6.4 Visualization

The Streamlit application will present localized risk information through Folium GIS maps, key metrics and Plotly line charts covering the 168-hour forward forecast horizon (Streamlit, n.d.; Plotly, n.d.; Folium, n.d.; Open-Meteo, n.d.).

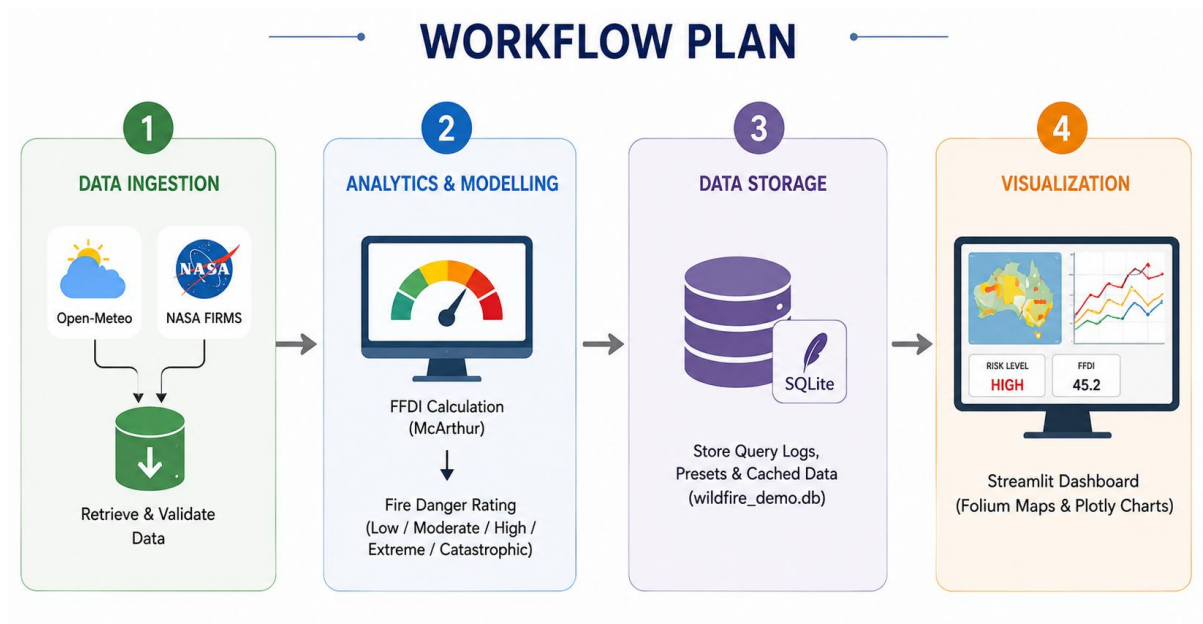


Figure 3 Workflow Plan

7 Task Allocation and Team Structure

The team will use a workstream-based structure. Responsibilities remain distinct, while integration, testing and major scope decisions will be reviewed collectively.

Table 1 Task Allocation and Team Structure

Member	Assigned Project Role	Responsibilities & Deliverables	Evidence
Member 1	Project Lead & Architect	Jira/GitHub setup (GitHub, n.d.), Draw.io architecture diagrams (draw.io, n.d.), report integration.	Jira board, GitHub, architecture
Member 2	Data Pipeline Engineer	Open-Meteo and NASA FIRMS API integration, data cleaning pipelines (NASA Fire Information for Resource Management System, n.d.).	Source code, tests, data documentation
Member 3	Database Lead	SQLite schema design for wildfire_demo.db and query-log implementation (SQLite, n.d.).	Schema, database code, documentation
Member 4	ML & Fire Physics Lead	McArthur FFDI implementation and spread-rate equations (Noble, Bary, & Gill, 1980).	Calculation module, validation results
Member 5	UI/UX	Streamlit layout, Folium maps and Plotly chart integration (Streamlit, n.d.; Folium, n.d.; Plotly, n.d.).	Dashboard, visualization code

8 Initial Risk Analysis

Table 2 Initial Risk Analysis

ID	Risk Description	Severity	Likelihood	Mitigation Strategy
R1	API rate limiting or outage during live demonstrations.	High	Medium	Implement local response caching in SQLite to store raw payloads.
R2	Inaccurate predictions caused by poor weather inputs.	Medium	Low	Validate humidity and wind inputs against boundary limits before calculations.
R3	Project scope creeps within the four-week delivery period.	High	Medium	Follow Jira sprint milestones and keep the database strictly locally using SQLite.
R4	Incorrect interpretation of fire-danger calculations.	High	Low	Validate formulas against the McArthur reference and document assumptions.
R5	Dashboard or GIS performance issues.	Medium	Medium	Cache data, limit map objects, and test representative geographic queries.
R6	Security exposure through API credentials.	High	Low	Use .streamlit/secrets.toml and exclude secrets from public version control using .gitignore.
R7	Missing out input parameters Drought Factor (DF) Calculation	High	High	Construct a derived analytical model that calculates the Drought Factor (DF) using 7-day precipitation and consecutive dry days.
R8	False Positive from Industrial Thermal Stats	Medium	Low	Remove low-confidence detections.
R9	Misaligned Task Allocation and	Medium	Medium	Use of Jira, MS Teams for communications and assigning tasks.

	Communication breakdown			
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9 Ethics, Privacy and Security Considerations

9.1 Ethical Safety and Disclaimers

Since fire predictions can influence public warnings and safety-related decisions (Plucinski, Sullivan, Rucinski, & Prakash, 2017), the system will provide transparent mathematical documentation for calculated indices. An explicit disclaimer will be displayed within the dashboard stating that the application is an academic decision-support prototype and is not a substitute for official state emergency services.

The project will avoid overstating model certainty. Values of forecasts and risk metrics will be provided in the proper context, and users will be advised to use official emergency information sources before making decisions.

9.2 Privacy and Security

- **Zero PII Collection:** The system processes spatial coordinates and meteorological data and does not collect or store any information about individual users (Guo & Li, 2025).
- **Open Data Compliance:** The NASA FIRMS and Open-Meteo data will be processed in compliance with their respective data use terms and licensing requirements (Data Use and Citation Guidance | NASA Earthdata, n.d.; Open-Meteo, n.d.)
- **Secret Management:** The API credentials will be stored in `.streamlit/secrets.toml` and kept outside of version control via `.gitignore` (Streamlit, n.d.).
- **Repository Security:** Credentials, local database files containing sensitive information and unnecessary private configuration will not be committed to GitHub.

10 Technology Stack

Table 3 Technology Stack

Layer	Technology	Purpose
Data acquisition	Python, Requests/async REST, Open-Meteo, NASA FIRMS	Retrieve weather forecasts and active fire observations.
Data processing	Pandas, NumPy	Validation, transformation, and feature preparation.
Fire analytics	Python mathematical model, McArthur FFDI	Calculate fire-danger indicators and derived outputs.
Storage	SQLite (wildfire_demo.db)	Query logs, regional presets, and caching.
GIS	Folium	Interactive geographic representation.
Dashboard	Streamlit	Interactive user interface.
Charts	Plotly	168-hour forecast and trend visualisation.
Development management	GitHub + Jira	Version control, task allocation, sprint planning, and evidence.

11 GitHub and Project Management Plan

The professional GitHub repository has been created to provide evidence of ongoing project development and will be maintained and updated throughout the semester. The repository will progressively include a README, assessment reports, architecture diagrams, workflow diagrams, source code and planning records. The repository is organised into a docs folder for reports and design documents, a diagrams folder for Draw.io exports, a src folder for application code, a data folder for data instructions and small permitted samples, a tests folder for validation, and a streamlit folder for local configuration files without any secrets.

Jira is used as the project management platform for the team. The board holds the task list, milestone planning, task assignment to all five members of the team (see Section 7), progress status and proof of completion, and is directly aligned with the initial system architecture in Section 5 and the workflow planning in Section 6. Git commits and pull requests will provide additional proof of collaboration throughout the development process.

Project Resources:

Table 4 GitHub and Project Management Plan

Resource	Link
GitHub Repository	<u>github.com/roshan-mahato/PRT_661_WILDFIRE_PREDICTION</u>
Jira Board	<u>Team Data Science Practice – Jira board</u>

12 Initial Project Timeline

Table 5 Initial Project Timeline

Week	Primary Activities	Expected Evidence
Week 1	Project setup, architecture, API exploration, Jira/GitHub setup.	Repository, Jira board, architecture diagram.
Week 2	API integration, validation, SQLite schema and FFDI implementation.	Working ingestion, database schema, calculation module.
Week 3	Streamlit dashboard, Folium GIS and Plotly forecast charts.	Interactive prototype and integration testing.
Week 4	Testing, risk review, documentation, report and demonstration preparation.	Final prototype, test evidence, report and updated repository.

13 Expected Outcomes

The project is expected to deliver a functional academic prototype that integrates real-time meteorological forecasts and satellite hotspot information into an interpretable fire-danger workflow. The key outputs will include FFDI-based risk indicators, localized ignition-related predictions, forward forecast visualisations, cached query records and an interactive GIS dashboard.

The project will also demonstrate professional data-science practice through version-controlled development, documented architecture, Jira-based task management, reproducible calculations, risk management and explicit ethical safeguards. Future enhancements could include machine-learning calibration of ignition probability, broader historical validation, automated deployment and additional environmental data sources.

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